Tutorial 8.

2. Exercise 1

Document 1 : D1 : "The Cat chased the Mouse."
Document 2 - D2 : "The dog Barked at the cat."
Document 3 - D3 : "The mouse ran away from the cat."

Step1 : Tokenization and Preprocessing

1. Convert all text to lowercase.

D1 : "the cat chased the mouse"
D2 : "the dog barked at the cat."
D3 : "the mouse ran away from the cat."

2. Remove punctuations - x

3. Tokenize each document into individual words (token)

D1 : "the cat chased the mouse"
D2 : "the dog barked at the cat."
D3 : "the mouse ran away from the cat."

4. Remove stop words.

D1 : "cat chased mouse"
D2 : "dog barked cat"
D3 : "mouse ran cat"

Step 2 : Vocabulary Construction.

List of tokens : ["barked", "cat", "chased", "dog", "mouse", "ran"]

Step3: Term frequency calculation.

vocab	D1	D2	D3
barked	0/3 : 0	1/3 : 0.33	0/3 : 0
cat	1/3 : 0.33	1/3 : 0.33	1/3 : 0.33
chased	1/3 : 0.33	0/3 : 0	0/3 : 0
dog	0/3 : 0	1/3 : 0.33	0/3 : 0
mouse	1/3 : 0.33	0/3 : 0	1/3 : 0.33
ran.	0/3 : 0	0/3 : 0	1/3 : 0.33.

Step 3.1: Normalized TF

$$w_{tf}(t,d) = \begin{cases} 1 + \log(tf_{t,d}) & \text{if } tf_{t,d} > 0 \\ 0 & \text{otherwise} \end{cases}$$

vocab	D1	D2	D3
barked	0	0.52	0
cat	0.52	0.52	0.52
chased	0.52	0	0
dog	0	0.52	0
mouse	0.52	0	0.52
ran	0	0	0.52

Step4 Inverse Document Frequency (IDF) calculation.

$$IDF(t) = \log\left(\frac{N}{1 + df_t}\right)$$

N = no. of document

df_t = no. of doc. where t appears

vocab	IDF
barked	$\log\left(\frac{3}{1+1}\right) = 0.176$
cat	$\log\left(\frac{3}{1+3}\right) = -0.124$
chased	$\log\left(\frac{3}{1+1}\right) = 0.176$
dog	$\log\left(\frac{3}{1+1}\right) = 0.176$
mouse.	$\log\left(\frac{3}{1+2}\right) = 0$
ran	$\log\left(\frac{3}{1+1}\right) = 0.176$

Step 5 Compute TF-IDF vectors.

$$TF\text{-}IDF(t,d) = TF_{t,d} \times IDF(t).$$

vocab	D1	D2	D3
barked	0	0.091	0
cat	-0.064	-0.064	-0.064
chased	0.091	0	0
dog	0	0.091	0
mouse	0	0	0
ran	0	0	0.091

3. "word2vec"

3.1. Exercice 2: Data Generation for CBOW.

D1: "The Cat chased the Mouse"

D2: "The dog Barked at the Cat."

D3: "The mouse ran away from the cat".

After stop-words removal:

D1: ["cat", "chased", "mouse"]

D2: ["dog", "barked", "cat"]

D3: ["mouse", "ran", "cat"]

Vocabulary:

["barked", "cat", "chased", "dog", "mouse", "ran"]

Indexing

Word	Index	One-Hot vector
barked	0	[1,0,0,0,0,0]
cat	1	[0,1,0,0,0,0]
chased	2	[0,0,1,0,0,0]
dog	3	[0,0,0,1,0,0]
mouse	4	[0,0,0,0,1,0]
ran	5	[0,0,0,0,0,1]

Step 2 - Look up table

For S1 : ["cat", "chased", "mouse"]

Context(words)	Context(index)	Target word	Target(Index)
chased	[2]	cat	1
cat, mouse	[1,4]	chased	2
chased	[2]	mouse	4

For S2 : ["dog", "barked", "cat"]

Context(words)	Context(index)	Target word	Target(Index)
barked	[0]	dog	3
dog, cat	[3,1]	barked	0
barked	[0]	cat	1

For S3 : ["mouse", "ran", "cat"]

Context(words)	Context(index)	Target word	Target(index)
ran	[5]	mouse	4
mouse, cat	[4,1]	ran	5
ran	[5]	cat	1